

ASSIGNMENT : 13 – WIRELESS TECHNOLOGIES

GUIDED BY : VIPUL MISTRI

Q.1 Scan for available Wi-Fi networks on your laptop or phone and list the SSIDs you find, noting which ones are open (unsecured) and which require a password.

ANS : Because I am an AI running on cloud servers, I don't have access to physical Wi-Fi hardware or antennas to scan local airwaves in real time.

However, if you want to perform a Wi-Fi scan on your own device using the command line, here is how you can do it:

On Windows (Command Prompt)

Open cmd as Administrator and run:

DOS

netsh wlan show networks mode=bssid

- **SSID:** The name of the Wi-Fi network.
- **Authentication:** Shows **Open** (unsecured) or **WPA2-Personal / WPA3-Personal** (requires a password).

On Linux (Terminal)

Run:

Bash

nmcli device wifi list

- Look under the **SSID** column for network names and the **SECURITY** column (-- means open/unsecured, while WPA2 / WPA3 means encrypted).

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Q.2 Set up a mobile hotspot on your phone, connect a laptop to it, and check what IP address your laptop receives from the hotspot. Hint: On Windows, use 'ipconfig' in Command Prompt; on Mac/Linux, use 'ifconfig' in Terminal.

ANS : To check the IP address assigned to your laptop while connected to your phone's mobile hotspot:

Step 1: Set Up and Connect

1. Turn on **Mobile Hotspot** on your smartphone.
2. Connect your laptop to your phone's Wi-Fi network.

Step 2: Find Your Assigned IP Address

On Windows (Command Prompt)

1. Press Win + R, type cmd, and press **Enter**.
2. Run the following command:

DOS

ipconfig

3. Look under the **Wireless LAN adapter Wi-Fi** section:
 - **IPv4 Address:** This is the local private IP address assigned to your laptop by your phone's hotspot DHCP server (typically in the 192.168.43.x range on Android or 172.20.10.x on iPhone)
 - **Default Gateway:** This is your phone's internal IP address handling the routing.

On Linux / macOS (Terminal)

1. Open your **Terminal**.
2. Run:

Bash

ip a

(On macOS, run ifconfig)

3. Find your active wireless interface (usually wlan0 on Linux or en0 on macOS) and check the inet value.

Q.3 Research and explain in your own words what WPA2 and WPA3 are, and list two differences between them relevant to security.

ASSIGNMENT : 13 – WIRELESS TECHNOLOGIES

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ANS : WPA2 (Wi-Fi Protected Access 2) and **WPA3** are Wi-Fi security protocols designed to authenticate devices on a wireless network and encrypt the data sent over the air. WPA2 has been the global standard since 2004, relying on a pre-shared key (PSK) and a 4-way handshake. WPA3 was introduced in 2018 as its successor to fix vulnerabilities inherent to WPA2 and upgrade wireless security for modern networks.

Key Differences Between WPA2 and WPA3

1. Protection Against Offline Password Guessing (Handshake Protocol)

- **WPA2:** Uses a **4-Way Handshake**. An attacker nearby can capture the handshake exchange over the air and attempt an offline brute-force or dictionary attack using specialized software to guess your password without interacting further with your router.
- **WPA3:** Replaces the 4-way handshake with **SAE (Simultaneous Authentication of Equals)**. SAE requires interacting with the router for every single authentication attempt, rendering offline password-cracking tools completely useless—even if your Wi-Fi password is short or simple.

2. Forward Secrecy

- **WPA2:** Does not support forward secrecy. If an attacker captures encrypted Wi-Fi traffic today and manages to steal or crack your Wi-Fi password years later, they can retroactively decrypt all the past network traffic they recorded.
- **WPA3:** Includes built-in **Forward Secrecy**. A new, unique session key is generated for every session. Even if someone eventually discovers your main Wi-Fi password, they cannot decrypt data intercepted prior to that moment.

ASSIGNMENT : 13 – WIRELESS TECHNOLOGIES

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Q.4 Find a news article or real-world case from the past 3 years where a public Wi-Fi network was used in a cyber attack, and summarize how the attack happened and what could have prevented it.

ANS : A major incident occurred when the **Australian Federal Police (AFP)** arrested and charged a 42-year-old Perth man for operating widespread **"Evil Twin" Wi-Fi attacks** targeting passengers at domestic airports and aboard Australian commercial flights.

How the Attack Happened

1. **Rogue Access Point Deployment:** The attacker deployed a portable wireless device to broadcast fake Wi-Fi access points that mimicked legitimate, trusted public networks (such as official airport or airline guest Wi-Fi networks).
2. **Captive Portal Cloning:** When unsuspecting travelers attempted to connect, they were redirected to a forged login page (captive portal).
3. **Credential Harvesting:** The rogue page prompted users to sign in using their personal email or social media accounts.
4. **Data Interception:** Once submitted, the attacker silently logged the user credentials and session tokens to his own storage, giving him unauthorized access to the victims' personal financial info, emails, and online services.

What Could Have Prevented It

For Individual Users (Clients)

- **Using a Mobile Hotspot / Cellular Data:** Avoiding public Wi-Fi entirely and connecting through cellular data (4G/5G) completely eliminates exposure to local rogue access points.

ASSIGNMENT : 13 – WIRELESS TECHNOLOGIES

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- **Connecting via a Virtual Private Network (VPN):** A VPN encrypts all traffic from the client device before it reaches the Wi-Fi radio, rendering intercepted data useless even on an untrusted or fake network.
- **Disabling Auto-Connect:** Disabling automatic Wi-Fi joining prevents phones from silently connecting to open networks broadcasting familiar SSIDs.
- **Inspecting Captive Portals:** Being cautious when an open network asks for email or social login credentials to access basic internet service.

For Venue Operators (Network Administrators)

- **Wireless Intrusion Prevention Systems (WIPS):** Modern enterprise Wi-Fi deployments use WIPS to actively detect and block rogue access points broadcasting identical SSIDs on the same radio frequencies.
- **802.11w (Management Frame Protection):** Enforcing MFP prevents malicious actors from spoofing deauthentication frames that force legitimate users off real networks and onto fake ones.